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I S-563b

APRIL 2, 1958

SUPERSEDING
Int. Fed. Spec. P–S–00563a (GSA–FSS)
August 22, 1956 and
Fed. Spec. P–S–563
October 25, 1949

FEDERAL SPECIFICATION

SOAP, BUILT, HIGH-TITER, POWDERED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers a powdered, high-titer, built soap that is suitable for high-temperature laundering of all types of wearing apparel and linens except silk and woolens.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

NN–K–231—Kegs, Wood, Slack.
UU–S–48—Sacks, Paper, Shipping.
PPP–B–41—Barrels; Wood, Slack.
PPP–D–723—Drums, Fiber.

Federal Standards:

Federal Standard No. 102—Preservation, Packaging, and Packing Levels.
Federal Standard No. 123—Marking for Domestic Shipment (Civilian Agencies).
Federal Test Method Standard No. 536—Soap and Soap Products (Including Synthetic Detergents); Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Composition.—The composition of the soap shall conform to the requirements shown in table I when tested as specified in 4.4.1.

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TABLE I.—*Composition*

Characteristics	Minimum	Maximum
	Percent	*Percent*
Moisture and matter volatile at 105° C.	---	16.0
Sum of free alkali and total matter insoluble in alcohol (alkaline salts)	40.0	---
Matter insoluble in water	---	1.0
Titer of mixed fatty acids prepared from the soap	39°C.	---
Material retained on a No. 12 sieve	---	1.5
Material passing a No. 140 sieve	---	20.0
Rosin acids (based on the finished product)	---	None
Anhydrous soap content	50.0	---
Free alkali, calculated as NaOH	---	16.0

3.1.1 *Computation.* — The percentage of matter volatile at 105° C. shall be calculated on the basis of the soap as received, but all other constituents shall be calculated on the basis of material containing 16.0 percent of matter volatile at 105° C. (for basis of purchase, see 6.2).

3.2 **Form.**—The soap shall be a uniform mixture of soap and alkalies in powdered form. It shall be readily soluble in hot water.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—For purposes of sampling, a lot shall consist of all soap manufactured as one batch; and offered for delivery at one time.

4.1.2 *Sampling for inspection of filled containers.*—A random sample of filled containers shall be selected from each lot offered for Government inspection in accordance with Military Standard MIL–STD–105 at inspection level I, and acceptable quality level (AQL) = 2.5 percent defective to verify compliance with all stipulations of this specification regarding fill, closure, marking and other requirements not involving tests.

4.1.3 *Sampling for tests.*—From each inspection lot offered for Government tests, two containers shall be selected at random. From each of the two containers enough soap to perform tests shall be taken and placed in separate clean, dry metal or glass containers, sealed, marked, and forwarded to the testing laboratory designated by the bureau or agency concerned.

4.2 Inspection.

4.2.1 *Inspection of filled containers.*—Each sample filled container selected in accordance with 4.1.2 shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample, having one or more defects, or under required fill, shall be rejected and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL–STD–105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests provided that the contractor has removed or repaired all nonconforming containers.

4.3 **Lot acceptance tests.** — The sample specimens selected in accordance with 4.1.3 shall be subjected separately to the tests specified in 4.4. If either specimen fails one or more of these tests, the lot shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or reworked all nonconforming products.

4.4 Test procedures.

4.4.1 *Composition.* — The composition of the soap shall be determined according to the methods in Federal Test Method Standard No. 536 as identified in table II.

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TABLE II.—*Test methods*

Determination	Method No.
Moisture and volatile matter (oven method)	201
Alcohol-insoluble matter	301
Free alkali as NaOH	302
Matter insoluble in water	303
Anhydrous soap	401
Rosin	701
Titer test	901
Sieve tests	2101

5. PREPARATION FOR DELIVERY

The definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packing.

5.1.1 *Level A.*—The soap shall be packed in quantities of 50, 75, 100, 200, or 300 pounds, as specified (see 6.1), in fiber drums (type II, grade A or type III), barrels (type II), kegs, or multiwall sacks (1W, 5W or 10W construction) conforming to Federal Specification PPP–D–723, PPP–B–41, NN–K–231 or UU–S–48, respectively. Slack barrels and kegs shall be provided with a two-way stretch creped kraft bag or liner having a minimum basis weight of 45 pounds per ream (24 × 36 – 500). The bag or liner shall be unwaxed and have a minimum stretch of 15 percent in each direction. Closure of the containers shall not be completed until the soap contents have cooled to room temperature.

5.1.2 *Level B.*—The soap shall be packed in quantities of 50, 75, 100, 200, or 300 pounds, as specified (see 6.1), in fiber drums (type I, grade A), barrels (type I), kegs or multiwall sacks (domestic type) conforming to Federal Specification PPP–D–723, PPP–B–41, NN–K–231, or UU–S–48, respectively. Slack barrels and kegs shall be provided with a two-way stretch creped kraft bag or liner having a minimum basis weight of 45 pounds per ream (24 × 36 – 500). The bag or liner shall be unwaxed and have a minimum stretch of 15 percent in each direction. Closure of the containers shall not be completed until the soap contents have cooled to room temperature.

5.1.3 *Level C.*—The soap shall be packed in containers in a manner to insure safe delivery and acceptance at destination. Containers shall comply with the Consolidated Freight Classification Rules or other carrier regulations applicable to the mode of transportation.

5.2 Marking.

5.2.1 *Civilian agencies.*—Shipping containers shall be marked in accordance with Federal Standard No. 123. Also, shipping containers shall be marked with the tare weight.

5.2.2 *Military agencies.* — In addition to any special marking specified in the contract or order, each shipping container shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data.—Purchasers should exercise any desired options offered herein (see 5.1 and 5.2).

6.2 Basis of purchase.—Built soap is subject to a possible gain or loss of weight, depending on atmospheric or storage conditions or both, or on packaging, as a result of fluctuation in the moisture content. Changes in the moisture content result in a corresponding change in the percentage of total solids or anhydrous soap content, or both.

6.2.1 Built soap should be purchased by net weight, provided the matter volatile at 105° C. is neither above nor below 13 percent.

6.2.2 Deliveries containing more than 16 percent of matter volatile at 105° C. should be rejected without further test.

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6.2.3 On deliveries containing 16 percent or less of matter volatile, at 105° C., settlement should be made on the basis of a product containing 13 percent moisture. The net weight of the material to be paid for on a 13 percent moisture and volatile basis is calculated as follows:

$$W = \frac{R \times (100-L)}{87}$$

where: W = Net weight of material to be paid for on 13 percent moisture and volatile basis.

R = Net weight of material as received.

L = Percentage loss at 105° C.

6.3 Transportation description.—Transportation description applicable to this item is:

Soap, not otherwise indexed by name, other than liquid.

Carload minimum weight 36,000 pounds.

Truckload minimum weight 36,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—Q M
Navy—Sh A
Air Force.

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